

Hamid Syed

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Applied AI, Computer Vision, and Robotics for real-world systems, with experience in autonomous platforms (UGV/UAV), reinforcement learning, and large scale data-driven deployment.

EDUCATION

Auburn University – (Alabama, USA)	<i>Aug 2026</i>
Ph.D. in Engineering (AI, Computer Vision & Robotics)	<i>GPA: 3.81/4.0</i>
Auburn University – (Alabama, USA)	<i>December 2025</i>
Master of Science in Data Science and Engineering	<i>GPA: 3.76/4.0</i>
Air University Islamabad (Islamabad, Pakistan)	<i>December 2022</i>
Master of Science in Electrical Engineering	<i>GPA: 3.52/4.0</i>
University of Engineering & Technology Peshawar (Peshawar, Pakistan)	<i>October 2020</i>
Bachelor of Science in Electrical Engineering	

ACADEMIC EXPERIENCE

Graduate Research Assistant – Auburn University	<i>May 2023 – Present</i>
- Designed end-to-end AI pipelines for autonomous inventory management - Developed unmanned ground vehicle (UGV) based perception systems - Creating drone-based detection systems. - Designing UAV and UGV based optimized spraying system	
Graduate Teaching Assistant (BSEN 3610) – Auburn University	<i>January 2025 – May 2025</i>
- Responsible for conducting laboratory experiments - Monitored instrumentation and control projects of undergraduates	
Graduate Research & Teaching Assistant – Air University Islamabad	<i>September 2020 – September 2022</i>
- Conducted research and published a conference paper - Conducted undergraduate laboratory lectures - Tutored and mentored undergraduate students	

PROFESSIONAL EXPERIENCE

AIOPs (Mobile, Alabama)	<i>May 2024 – August 2024</i>
<i>Machine Learning Intern</i> - Research and Development - Reinforcement learning control of real-world industrial systems - Research on Proximal Policy Optimization and Multi-Agent Learning	
Datapoint – Airloop (Islamabad, Pakistan)	<i>October 2022 – April 2023</i>
<i>AI Engineer</i> - Designed pavement crack detection and applied Geo mapping - Developed commercial-level Traffic Analytics model - Managed and supervised the AI team - Developed a prototype for UAV-based pavement crack detection system	
<i>AI Engineer Intern</i> Developed a prototype for UAV-based pavement crack detection system	
Pakhtunkhwa Energy Development Organization (Swat, Pakistan)	<i>June 2018 – July 2018</i>
<i>Electrical Engineer Intern</i> Understanding the design and construction of “run of the river” dam (36.6MW)	

SCHOLARLY WORK

- Reviewer, Sensors International
 - Judge for the Capstone Design Research showcase (Auburn - 2024)
 - Declared and Certified as a [Member](#) of Alpha Epsilon (The Honor Society of Biological Engineering)
 - Attended and presented at the American Society of Agricultural and Biological Engineers (ASABE) annual meeting in Nebraska-2023, and Los Angeles, California-2024.
 - Attended and presented at the International Conference on Precision Agriculture (ICPA) 2024, Kansas, USA.
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PUBLICATIONS

1. **Syed Hamid** and M H Mahmood, “3D Human Reconstruction with Corresponding 3D Texture Model: A Comparison of Salient Approaches” (ICETEC - 2022)
 2. MH Rahman, **Syed Hamid**, and Rehman et al., “Forecasting Growth Dynamics of Hydroponic Kale in Controlled Environment Agriculture through Vision-Based Phenotyping and Time-Series Modeling” (Smart Agricultural Technology - 2025)
 3. **Syed Hamid et al.**, “Automated in-field inventory management and quality assessment system using end-to-end deep learning for ornamental nursery crops”, (Smart Agricultural Technology - Under Review)
 4. **Syed Hamid et al.**, “Exploring 3D Latent Feature Embedding for Few and Single-Shot Unsupervised Person Re-Identification (Sensors International - Under Review)
 5. Ahmad, Faraz, **Syed Hamid**, and Rehman et al., “Autonomous Robotic Navigation in Ornamental Crop Production Using Vision and Sensor Fusion” (Computer Electronics in Agriculture - Under Review)
 6. Rafi, Mohtasim, **Syed Hamid**, and Rehman et al., “Cloud-Enabled Plant Segmentation and Tracking Framework for Ornamental Nursery Inventory Management” (Computer Electronics in Agriculture - Under Review)
 7. Waseem, Muhammad, **Syed Hamid**, and Rehman et al., “Transforming Agriculture with Cyber-Physical Systems: An Insight into Future Smart Farming” (Smart Agricultural Technology - Under Review)
 8. Waseem, Muhammad, **Syed Hamid**, and Rehman et al., “Generative AI-Based Digital Twin Pipeline for Advanced Agricultural Analytics Enabled by Synthetic Data and Zero-Shot Learning” (Computer Electronics in Agriculture - Under Review)
 9. Book Chapter: Ahmad Fayaz, Ruihong Zhang, and **Syed Hamid et al.** “The Art of Doing Research” (Under Review)
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PRESENTATIONS

Oral Presentation

- **Syed Hamid.** & Rehman, T. U. (2024). “Smart Inventory Management and Quality Assurance System for Ornamental Plants”. Auburn Research Symposium, Auburn, Alabama, USA. March 26, 2024. ([WINNER](#))
- **Syed Hamid et al.**, “Proactive Resource Efficiency via Coordinated Imaging and Sprayer Execution (*PRECISE*)”. NASA AgAir Competition, USA. May 21, 2025. ([FINALIST](#))

Poster Presentation

- **Syed Hamid.** & Rehman, T. U. (2024). “A label-free deep learning approach for ornamental nursery inventory management”. Data Science and AI in Production Agriculture, Auburn, Alabama, USA. April 26, 2025. (**WINNER**)
 - **Syed Hamid.** & Rehman, T. U. (2023). “Automated In-field Ornamental Nursery Plant Counting and Quality Assessment With End-to-End Deep Learning for Inventory Management”. College of Agriculture Graduate Research Poster Showcase, Auburn, Alabama, USA. October 26, 2023. (**WINNER**)
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GRANTS & FUNDED RESEARCH

- Awarded **\$4,966** through the Alabama Agriculture Experimental Station’s Professional Enhancement for Emerging Researchers (**PEER**) Grant Program for the project “*Hybrid Aerial-Ground Robotic Vision System for Precision Inventory and Quality Assessment in Ornamental Nurseries*”.
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AWARDS & GRANTS

- [Finalist in NASA AgAir Competition](#) and a stipend of \$9000. (May 2025)
 - Winner of Poster Competition at “Data Science and AI in Production Agriculture” and a cash prize of \$500 (March 2025)
 - [Excellence in Productivity Award](#) at the Farm Robotics Challenge at FIRA - World Ag Robotics Forum and a cash prize of \$5000 (October 2024)
 - [Recipient of FFAR Fellowship](#) (August 2024)
 - Nominated for Graduate School Merriwether Fellowship Award (Auburn – 2024)
 - Winner of Auburn Research Symposium Oral Presentation(Auburn - March 2024)
 - Winner of Graduate Student Poster Competition (Auburn - Nov 2023)
 - [Winner of National Idea Bank Competition](#) (Natural Resources Sector – 2021)
 - Merit based full fee waiver (2020 – 2022)
 - Recipient of National Talent Scholarship (2016 – 2020)
 - Merit based tuition fee waiver (2014 – 2016)
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SKILLS

- Programming Languages: Python, MATLAB, R, C++, LaTeX
 - Libraries and Frameworks: OpenCV, NumPy, Pandas, Matplotlib, Scikit-learn, object detection API
 - Design: AutoCAD, Proteus, PSpice
 - Developer Tools: Linux, Docker, JupyterLab, Pycharm, Visual Studio Code
 - Hardware: RTX 5090, Jetson Orin, Micro Controller 8051, Esp 702 Micro Controller
 - Photogrammetry and Reconstruction: NVIDIA Omniverse, NuRec, Pix4DMapper, Pix4d Fields, DJI Terra, Isaac Sim, Correlator3D, Agisoft, ArcGIS Pro,
 - Experienced with ROS2, small UAVs, large UAVs, unmanned ground vehicles as well as their applications
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CERTIFICATIONS

- Certified UAV Remote Pilot (Federal Aviation Association)
 - Pesticide Applicator Permit (Alabama Department of Agriculture & Industries)
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PROJECTS

Ornamental Nursery Inventory Management System (Ph.D. Research Project)

- Developing an Unmanned Ground Vehicle based data collection and analysis pipeline
- Developing deep learning and statistics-based model for plant counting and quality assessment
- Integrating NVIDIA Omniverse and NuRec for generating synthetic nursery environments

Unmanned Aerial Vehicle (UAV) based Weed Detection System (Ph.D. Research Project)

- Data collection and analysis using different UAVs (Phantom4M and Mavic3M)
- Build a digital twin pipeline to generate synthetic datasets for training agricultural vision systems
- Generating photogrammetry and processing multispectral imagery using deep learning techniques

UAV and UGV based Smart Spraying System (Ph.D. Research Project)

- Using camera modalities and different software for spray path generation
- Smart Spraying by utilizing embedded systems and deep learning-based object segmentation models.

Development of Deep Learning based Pavement Condition Index, Survey, GIS mapping, and reporting of 5000 kms of roads in Khyber Pakhtunkhwa. (Corporate Project - 2023)

- Development of a deep learning-based pavement condition index (PCI) for assessing the condition of roads
- Use of GIS mapping technology to analyze and visualize the collected data, allowing for efficient identification of problematic areas and prioritization of maintenance and repair efforts.

Deep Learning Based Road Traffic Analytics system based on axle-load. (Corporate Project - 2023)

- Collection and annotation of a diverse novel dataset, which consists of 33 different classes
- Implementation of the system at strategic locations, such as toll booths or weigh stations, to collect data on vehicle traffic and ensure compliance with weight limits

Exploring a 3D Latent Feature Space Embedding, for Person Re-identification (MS Thesis Project - 2022)

- Explored 3D latent Space, transformed 2D images into 3D Images embedding, and utilized it for person Re-Identification.
 - Proposed a Novel Method for 3D Person Re-Identification that mimics the Human Brain functionality.
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